MEDIUM   ALGEBRA

Algebra

15

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ALGEBRA

$$8x + y = 5$$

$$y = 9x + 1$$

The solution to the given system of equations is x, y. What is the value of x?

A. -6

B. $\frac{4}{17}$

C. $\frac{6}{17}$

D. 4

Q2 ALGEBRA

A neighborhood consists of a 2-hectare park and a 35-hectare residential area. The total number of trees in the neighborhood is 3,934. The equation $2x + 35y = 3{,}934$ represents this situation. Which of the following is the best interpretation of x in this context?

A. The average number of trees per hectare in the park

B. The average number of trees per hectare in the residential area

C. The total number of trees in the park

D. The total number of trees in the residential area

Q3 ALGEBRA

The function $f x = 55.20 - 0.16x$ gives the estimated surface water temperature $f x$, in degrees Celsius, of a body of water on the xth day of the year, where $220 \leq x \leq 360$. Based on the model, what is the estimated surface water temperature, in degrees Celsius, of this body of water on the 326th day of the year?

A. 55.20

B. 3.04

C. -0.16

D. -52.16

Q4 ALGEBRA

A company that provides whale-watching tours takes groups of 21 people at a time. The company's revenue is 80 dollars per adult and 60 dollars per child. If the company's revenue for one group consisting of adults and children was 1,440 dollars, how many people in the group were children?

A. 3

B. 9

C. 12

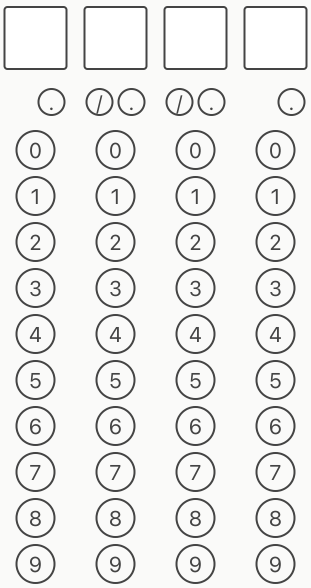
D. 18

Q5

GRID-IN ALGEBRA

A line passes through the points $4, 6$ and $15, 24$ in the xy-plane. What is the slope of the line?

STUDENT-PRODUCED RESPONSE



Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6 ALGEBRA

The boiling point of water at sea level is 212 degrees Fahrenheit (°F). For every 550 feet above sea level, the boiling point of water is lowered by about 1°F. Which of the following equations can be used to find the boiling point B of water, in °F, x feet above sea level?

A. $B = 550 + \frac{x}{212}$

B. $B = 550 - \frac{x}{212}$

C. $B = 212 + \frac{x}{550}$

D. $B = 212 - \frac{x}{550}$

Q7 ALGEBRA

$$y = 6x + 16$$

$$-7x - y = 36$$

What is the solution x, y to the given system of equations?

A. $-4, -8$

B. $-\frac{20}{13}, -\frac{80}{13}$

C. $4, 40$

D. $20, 136$

Q8

ALGEBRA

x	1	2	3
y	11	16	21

The table shows three values of x and their corresponding values of y. Which equation represents the linear relationship between x and y?

A. $y = 5x + 6$

B. $y = 5x + 11$

C. $y = 6x + 5$

D. $y = 6x + 11$

Q9

GRID-IN ALGEBRA

A linear model estimates the population of a city from 1991 to 2015. The model estimates the population was 57 thousand in 1991, 224 thousand in 2011, and x thousand in 2015. To the nearest whole number, what is the value of x?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10 ALGEBRA

Which of the following systems of linear equations has no solution?

A. $x = 3$
$y = 5$

B. $y = 6x + 6$
$y = 5x + 6$

C. $y = 16x + 3$
$y = 16x + 19$

D. $y = 5$
$y = 5x + 5$